Concern Resolution Process for In-house Manufacturing.

QC Story.

QC STORY PROJECT

To Eliminate Quality Concern of Cab & Cargo Mismatch Issue on DC Scorpio Vehicle.

Mahindra and Mahindra Ltd, Scorpio Body Shop, Nasik Plant-1

	Plan	Actual
Start Date	**3rd May-17**	**4th May-17**
Completion Date	**30th Nov-17**	**29th Nov-17**

Current Status: Closed.

Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 2: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardise
- Step 9: Summarise & Plan future actions
- Guidelines to user

Cross Functional Team

Team Leader – Leads the team, Co-ordinates the meeting
Mentor - Facilitates the team for concern resolution, provides resource.
Mostly Mentor is PU Manager.
Members – Cross Functional Team

No	Description	Name	Designation	Dept	Phone	Email
1	**Team Leader**	**N.S.Hiray**	Deputy Manager	QA, SBS	9881737077	Hiray.nitin2@mahindra.com
2	**Mentor**	**Prasad Upasani**	Manager	QA, SBS	9881232185	Upasani.Prasad@mahindra.com
3	**Member 1**	**Sudipta Ghosal**	Manager	Prod. SBS	9922932354	Ghosal.Sudipta@mahindra.com
4	**Member 2**	**Sushil Dwivedi**	Asst. Manager	Tooling SBS	9922971514	Dwivedi.sushil@mahindra.com
5	**Member 3**	**Santosh Jadhav**	Deputy Manager	MQA. TCF	9850848982	Jadhav.santosh@mahindra.com

1. Select the Theme

Brief Description :
To Eliminate the Quality Concern of Cab & Cargo Mismatch Issue on DC Scorpio Vehicle.

What: **Double Cab & Cargo not in line issue at C pillar Area on DC Vehicle.**

Where: **Cargo-Cab mismatch issue in TCF.**

Who: **Concern Reported from Field Customer (Chile) & CAI {TCF}.**

When: **Since Jan-17.**

Why: **To improve STR & ICC PPM.**

How: **Total Avg.30 Defects/ Month.**

Cab & Cargo Mismatch issue from Scorpio Double Cab Model on STCF Line Since Jan-17, Chronic V1 Concern leading to STR & ICC PPM, Reported from Field Customer & CAI, Total Avg 30 Vehicles/Month.

1. Select the Theme

Brief Description of Concern

- **In TCF, After fitment of Cargo on Chassis, Cargo-Cab mismatch issue observed on DC Scorpio Vehicle.**

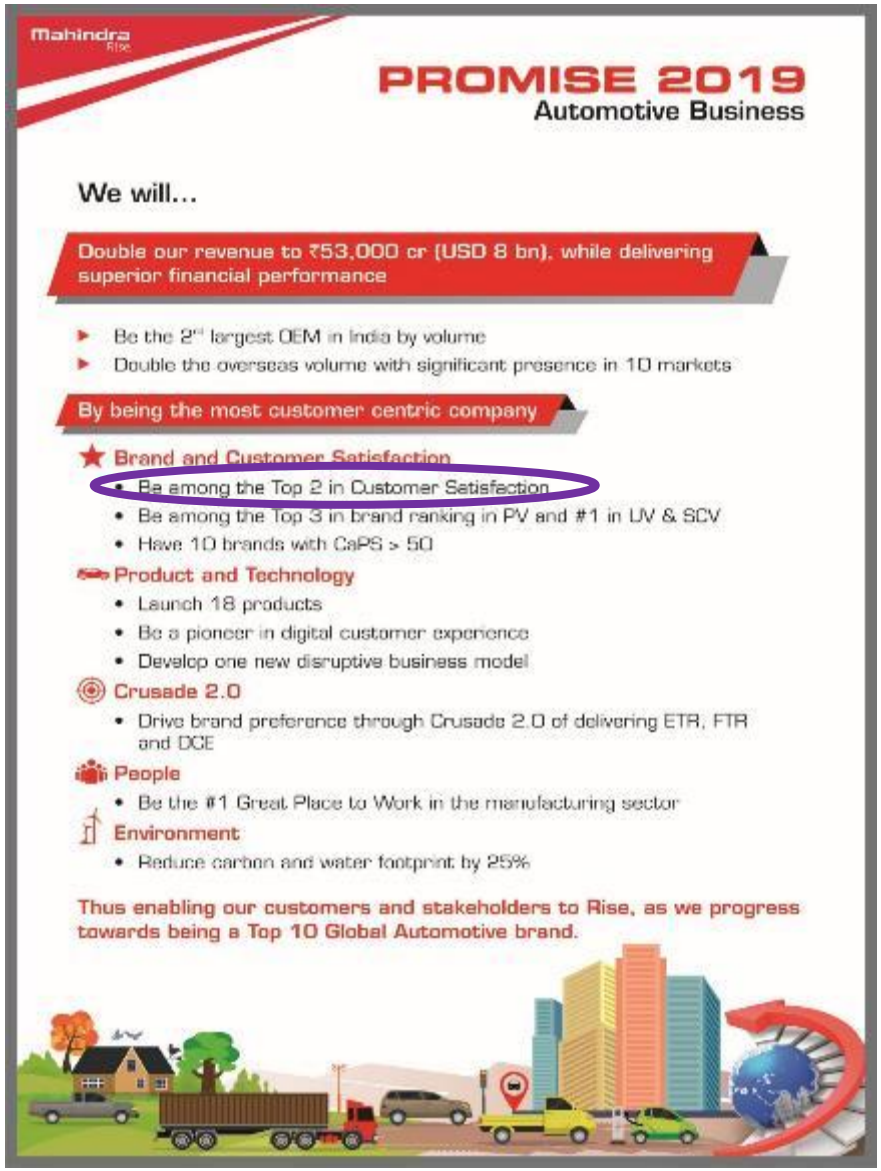
- **Chronic Concern Reported by internal customer/ TCF (CAI) on Double Cab Scorpio Vehicle.**

- **Major STR Concern in TCF.**

1. Select the Theme

Brief Description of Concern

Eliminating Cab & Cargo Mismatch Issue on DC Scorpio Vehicle.



Mahindra Rise

PROMISE 2019
Automotive Business

We will...

Double our revenue to ₹53,000 cr (USD 8 bn), while delivering superior financial performance

- Be the 2nd largest OEM in India by volume
- Double the overseas volume with significant presence in 10 markets

By being the most customer centric company

Brand and Customer Satisfaction
- Be among the Top 2 in Customer Satisfaction
- Be among the Top 3 in brand ranking in PV and #1 in UV & SCV
- Have 10 brands with CaPS > 50

Product and Technology
- Launch 18 products
- Be a pioneer in digital customer experience
- Develop one new disruptive business model

Crusade 2.0
- Drive brand preference through Crusade 2.0 of delivering ETR, FTR and DCE

People
- Be the #1 Great Place to Work in the manufacturing sector

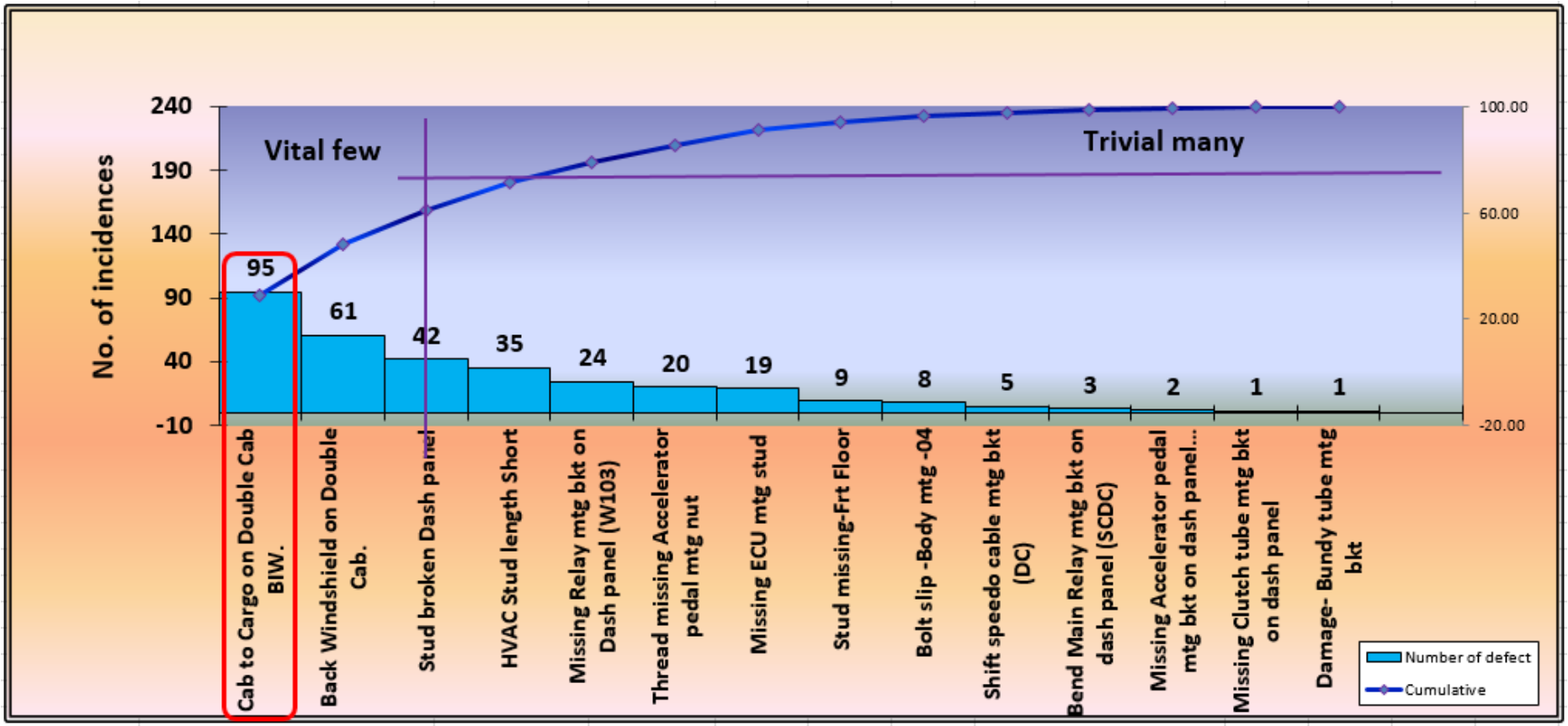
Environment
- Reduce carbon and water footprint by 25%

Thus enabling our customers and stakeholders to Rise, as we progress towards being a Top 10 Global Automotive brand.

2. Justify the choice (reasons)

Pareto Chart

TCF Defect Data Feb-17 to April-17 (Commodity –BIW).



Cab & Cargo Mismatch Issue is the Top Concern of TCF.

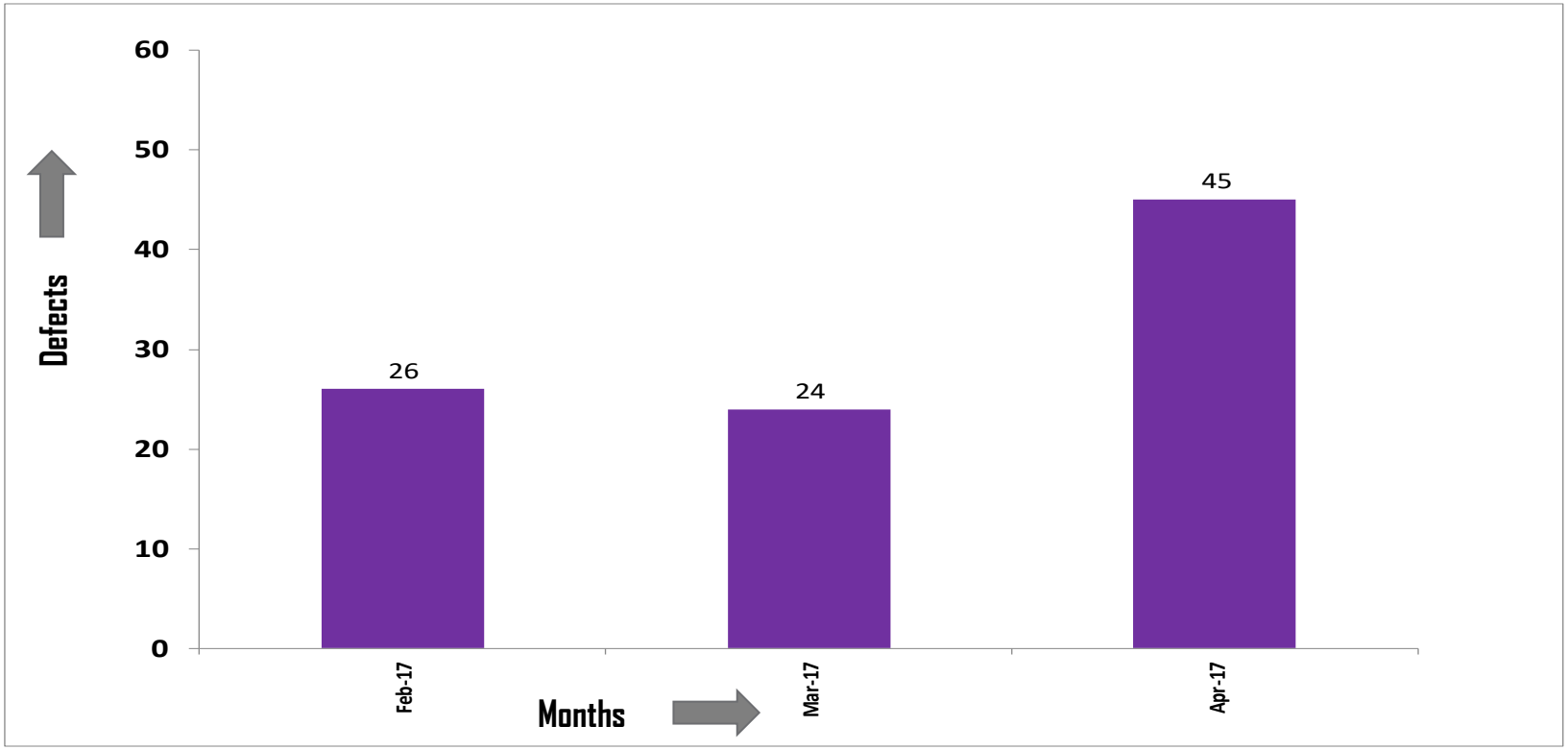
2. Justify the choice (reasons)

D. COPQ (Cost of Poor Quality) Calculation

Description	Feb-17 to April-17
Number of pieces rejected last month (for the part number identified for study)	95
Number of pieces(Vehicle) reworked last month	95
Rework man hours Cost Including (Labor Cost+ FOS+ Electricity) in SAP.	758 ` / hr.
Time required to rework the defect.	15 min Avg.
Total time required to rework the defects (in mins.)	1425 Mins.
Total time required to rework the defects (in Hrs.)	23.75 Hrs.
Total rework cost(Rs.) for Feb-17 to April-17 month	18002.50 `

2. Justify the choice (reasons)

CAI Data from Feb-17 to April-17.

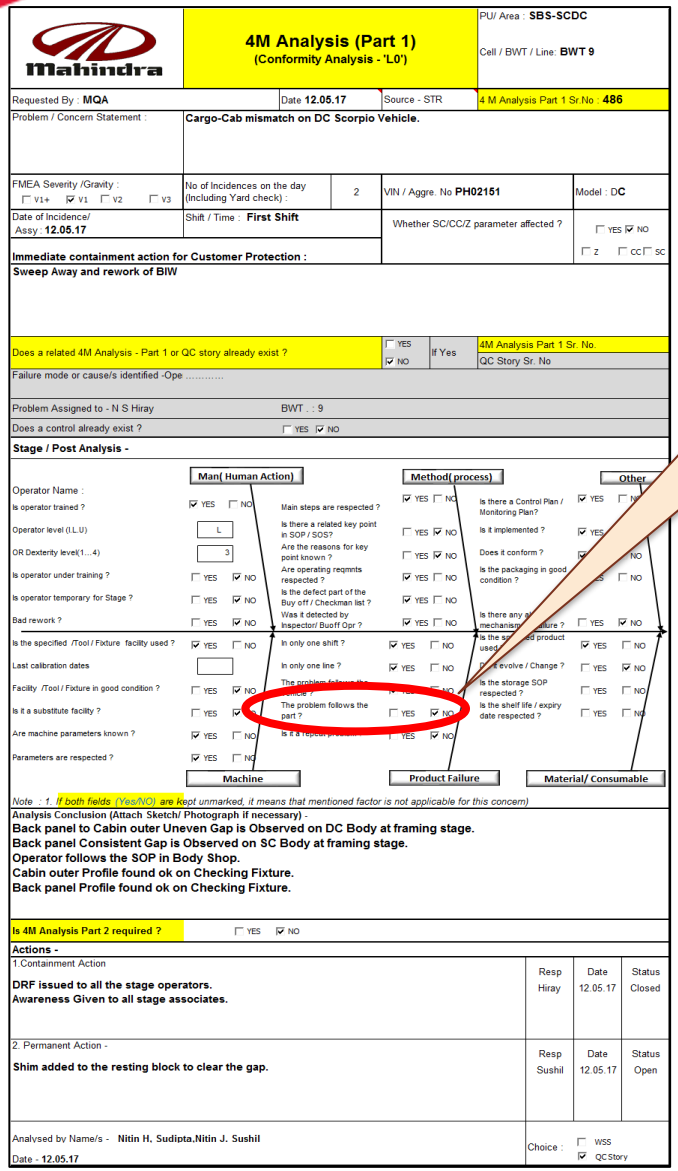


Last 3 Month Data – 95 Defects of Cargo-Cab mismatch on DC Scorpio Vehicle.

2.a Immediate Containment Action

- **100% Rework of Defective Double Cab BIW started at CAI stage(TCF).**

3. UNDERSTAND THE CURRENT SITUATION



4M Analysis (Part 1)
(Conformity Analysis - 'L0')

PU/ Area : SBS-SCDC
Cell / BWT / Line: BWT 9

Requested By : MQA | Date 12.05.17 | Source - STR | 4 M Analysis Part 1 Sr.No : 486

Problem / Concern Statement : Cargo-Cab mismatch on DC Scorpio Vehicle.

FMEA Severity /Gravity : ☐ V1+ ☑ V1 ☐ V2 ☐ V3 | No of Incidences on the day (Including Yard check) : 2 | VIN / Aggre. No PH02151 | Model : DC

Date of Incidence/ Assy : 12.05.17 | Shift / Time : First Shift | Whether SC/CC/Z parameter affected ? ☐ YES ☑ NO ☐ Z ☐ CC ☐ SC

Immediate containment action for Customer Protection :
Sweep Away and rework of BIW

Does a related 4M Analysis - Part 1 or QC story already exist ? ☐ YES ☑ NO | If Yes | 4M Analysis Part 1 Sr. No. | QC Story Sr. No

Failure mode or cause/s identified -Ope .....

Problem Assigned to - N S Hiray | BWT . : 9

Does a control already exist ? ☐ YES ☑ NO

Stage / Post Analysis -

Man(Human Action)		Method(process)		Other	
Operator Name :		Main steps are respected ?	☑ YES ☐ NO	Is there a Control Plan / Monitoring Plan?	☑ YES ☐ NO
Is operator trained ?	☑ YES ☐ NO	Is there a related key point in SOP / SOS?	☐ YES ☑ NO	Is it implemented ?	☑ YES
Operator level (I.L.U)	L	Are the reasons for key point known ?	☐ YES ☑ NO	Does it conform ?	NO
OR Dexterity level(1...4)	3	Are operating reqmnts respected ?	☑ YES ☐ NO	Is the packaging in good condition ?	NO
Is operator under training ?	☐ YES ☑ NO	Is the defect part of the Buy off / Checkman list ?	☑ YES ☐ NO		
Is operator temporary for Stage ?	☐ YES ☑ NO	Was it detected by inspector/ Buoff Opr ?	☑ YES ☐ NO	Is there any [illegible] mechanism [illegible]ure ?	☐ YES ☑ NO
Bad rework ?	☐ YES ☑ NO			Is the [illegible]ed product used	☑ YES ☐ NO
Is the specified /Tool / Fixture facility used ?	☑ YES ☐ NO	in only one shift ?	☑ YES ☐ NO	Does it evolve / Change ?	☐ YES ☑ NO
Last calibration dates		In only one line ?	☑ YES ☐ NO	Is the storage SOP respected ?	☐ YES ☐ NO
Facility /Tool / Fixture in good condition ?	☐ YES ☑ NO	The problem follows the [illegible]	[illegible]	Is the shelf life / expiry date respected ?	☐ YES ☐ NO
Is it a substitute facility ?	☐ YES ☑	The problem follows the part ?	☐ YES ☑ NO		
Are machine parameters known ?	☑ YES ☐ NO	Is it a repair [illegible]	☐ YES ☑ NO		
Parameters are respected ?	☑ YES ☐ NO				
Machine		**Product Failure**		**Material/ Consumable**	

Note : 1. If both fields (Yes/NO) are kept unmarked, it means that mentioned factor is not applicable for this concern)

Analysis Conclusion (Attach Sketch/ Photograph if necessary) -
Back panel to Cabin outer Uneven Gap is Observed on DC Body at framing stage.
Back panel Consistent Gap is Observed on SC Body at framing stage.
Operator follows the SOP in Body Shop.
Cabin outer Profile found ok on Checking Fixture.
Back panel Profile found ok on Checking Fixture.

Is 4M Analysis Part 2 required ? ☐ YES ☑ NO

Actions -

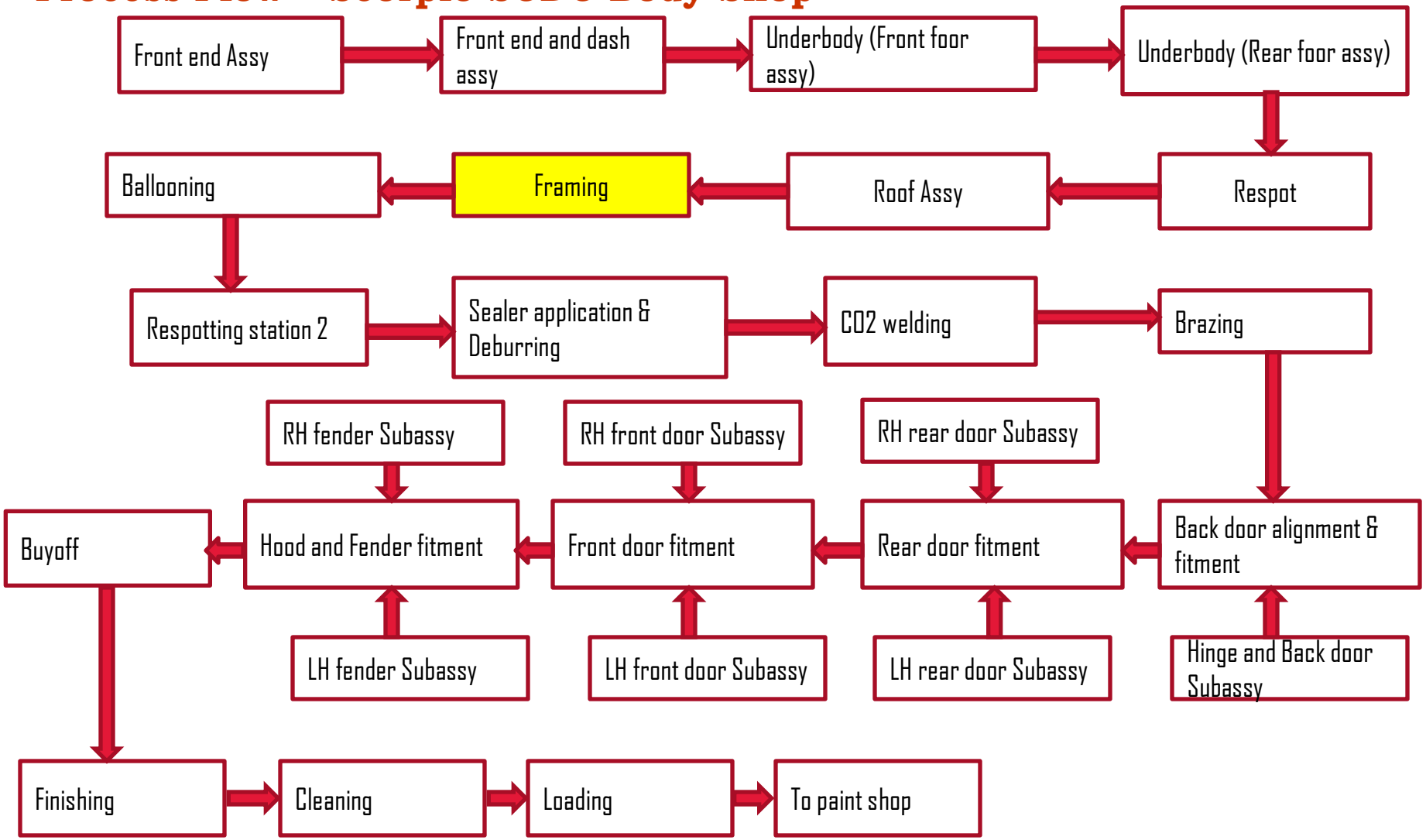
	Resp	Date	Status
1.Containment Action DRF issued to all the stage operators. Awareness Given to all stage associates.	Hiray	12.05.17	Closed
2. Permanent Action - Shim added to the resting block to clear the gap.	Sushil	12.05.17	Open

Analysed by Name/s - Nitin H, Sudipta,Nitin J, Sushil
Date - 12.05.17
Choice : ☐ WSS ☑ QC Story

Inference from L0.

4M analysis. Good and not so good observation	
MAN	Associate is Dext. Level 4 and Black L
METHOD	**Associate aware of the key point and after body & Cargo Mth tightening, mismatch issue occurred.**
OTHERS	Control plan adhered on stage
MACHINE	Tool used are calibrated
PRODUCT	**Problem follows the part**
MATERIAL	Handling of cab & Cargo is no issue

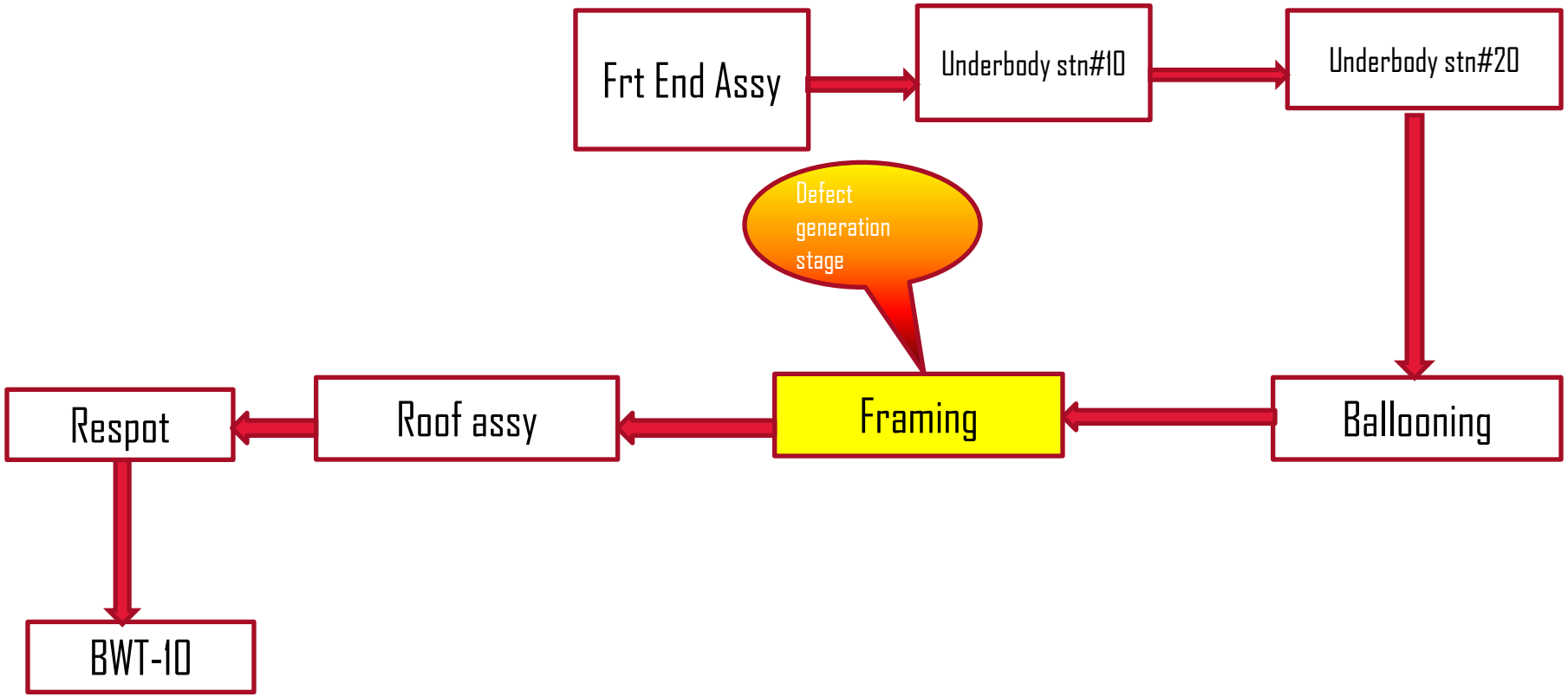
Process Flow – Scorpio SCDC Body Shop



Note: Parts shaded in Yellow are Defect Generated Stage.

3. Understand the current situation

Observation of Process at Framing Line (BWT-09)

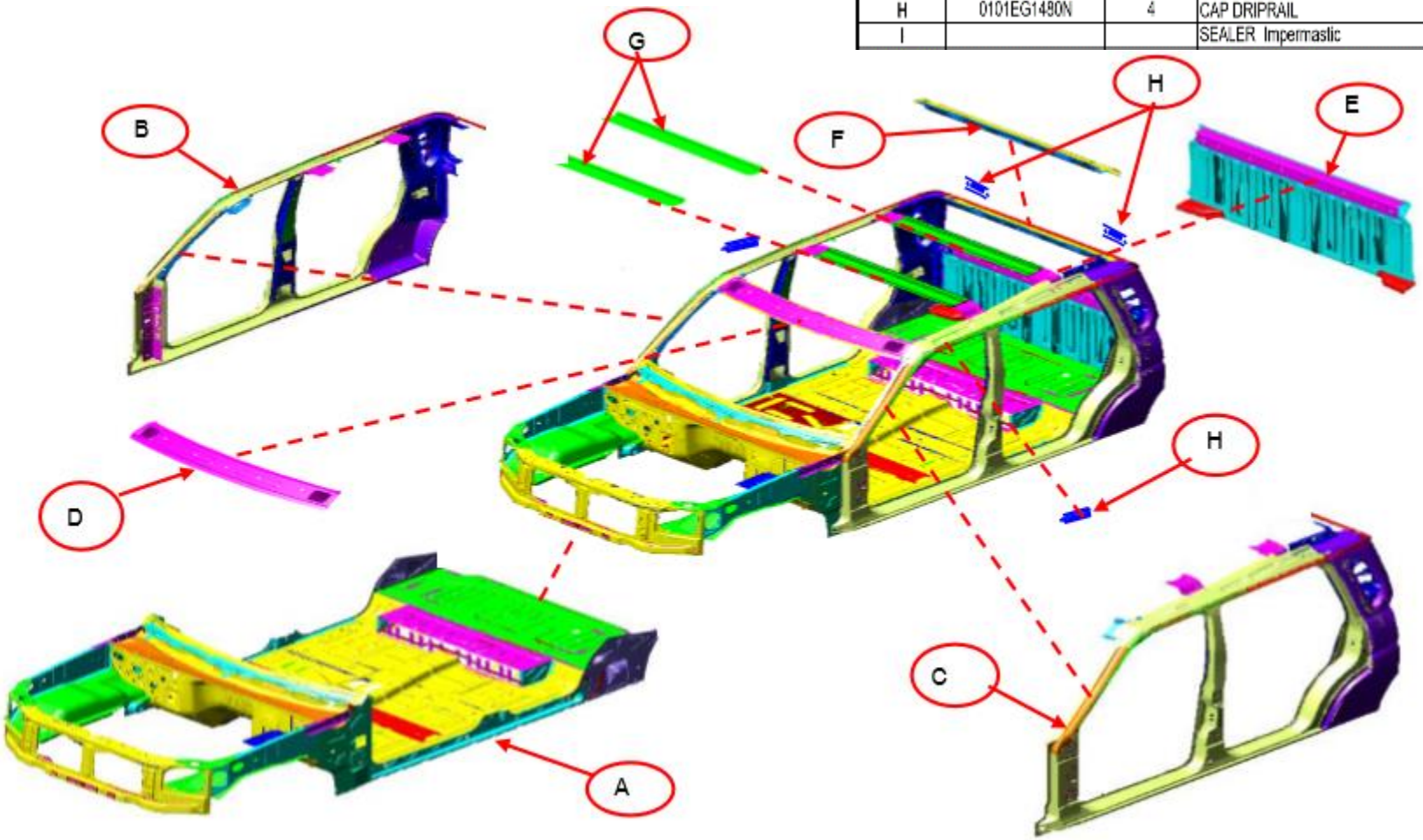


Note: Parts shaded in Yellow are Defect Generated Stage.

3. Understand the Current Situation

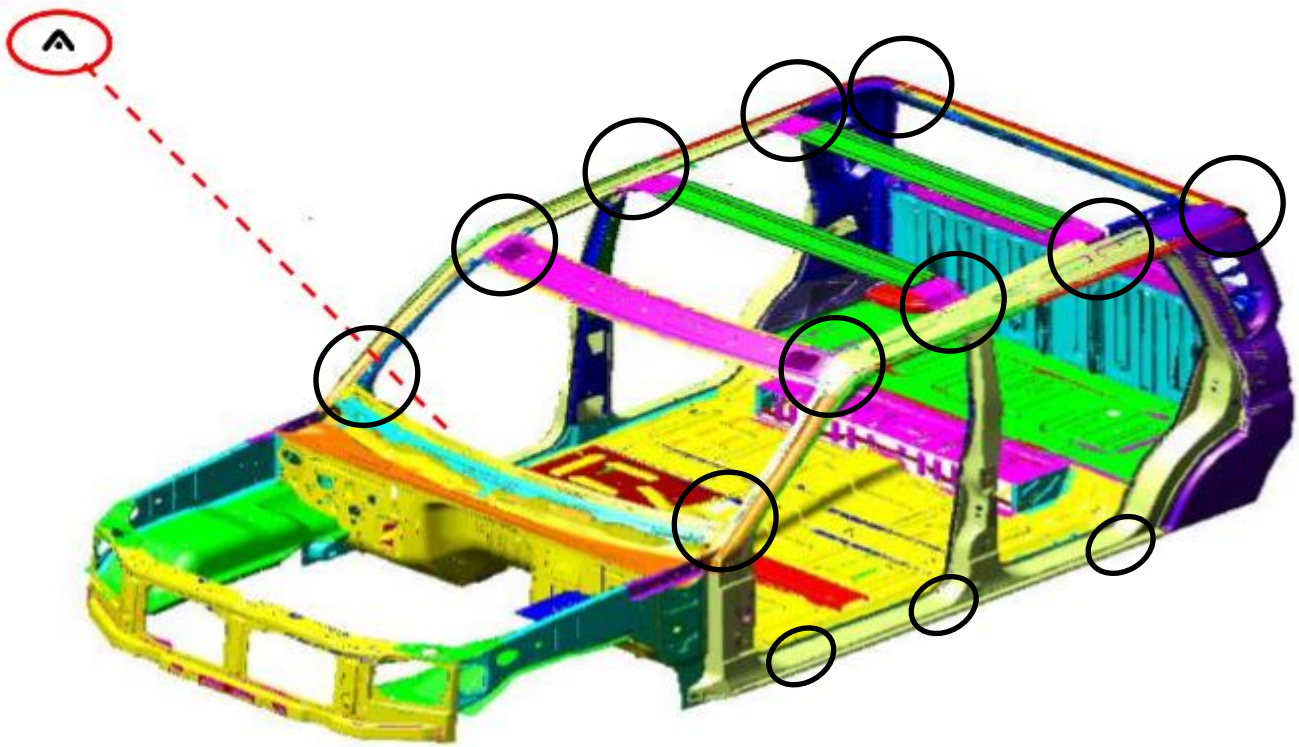
Pre-Defect Generated Stage
Pictorial View – Ballooning Stage

Sequence	Part No	Qty/Assy	Part Description
A	NA	1	U/BODY ASSY DC
B	0101EL0200N	1	SIDE ASSY DC RH
C	0101EL0210N	1	SIDE ASSY DC LH
D	0101DG0010N	1	PNL ASSY HEADER FRT
E	0101GL0090N	1	ASSY BACK PANEL DC
F	0101DL0160N	1	PNL ASSY HEADER REAR DC
G	0101DG0030N	2	ROOF BOW FRONT DC
H	0101EG1480N	4	CAP DRIPRAIL
I			SEALER Impermastic



3. Understand the current situation

Defect Generated Stage
Pictorial View – Framing Stage



- **Gun Spot Welding operation done at Frt Header, 1st & 2nd Roof Bow, Rear Header, Plenum, Rocker Panel, Back Panel Joinery Area.**

3. Understand the current situation

Data collection for Double Cab Cargo-Cab mismatch issue.

	Dates																				
	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Ist	2	3	2	1	3	2	1		3	2	1	2	2	2		2	3	1	3	2	2

During Data collection, we observed the following:

1) **Defect Observed on Both Side i.e. Left & Right Corner of DC Vehicle.**

2) **Single Cab Vehicle Having Zero Cargo-Cab mismatch issue.**

3. Understand the current situation

Process Walk Through – Scorpio Body Shop.

1. Back panel to Cabin outer Uneven Gap is Observed on DC Body at framing stage.
2. Back panel Consistent Gap is Observed on SC Body at framing stage.
3. Operator follows the SOP in Body Shop.
4. Cabin outer Profile found ok on Checking Fixture.
5. Back panel Profile found ok on Checking Fixture.

Process Walk Through –TCF (Trim-03).

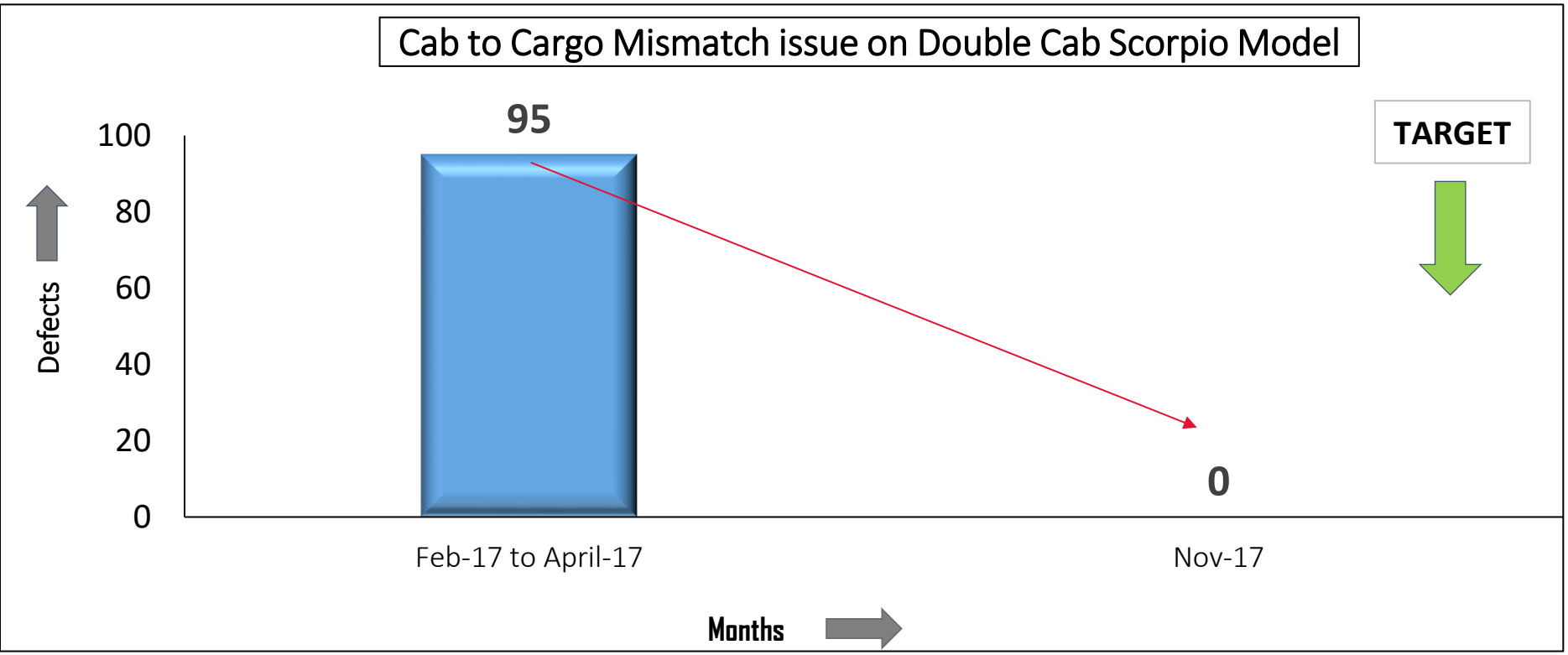
1. Cargo & Cab having Common width/Dimensions for Single Cab & Double Cab Model.
2. Operator follows the SOP in TCF.
3. Cargo-Cab mismatch issue Observed on DC Vehicle at middle section Either Left Side/Right Side or Both Side.
4. Final inspection is done at CAI and if found Not Ok, Rework was done.
5. After fitment of Cargo on Chassis, Cargo-Cab mismatch issue observed on Final Line in TCF.

Quick Win Implementation :

- **No Quick Wins applicable for Cab to Cargo Mismatch issue.**

4. Select Target

Present Level :	**Total 95 vehicles reported for the defect from Feb-17 to April-17.**
Target :	**Zero Defect (30th Oct-17)**



5. Analysis

5.1 Brain Storming

Brain Storming Team

Nitin Hiray.	Pravin Patil
Prasad Upasani	Ganesh Bendale
Sudipta Ghosal	Sheetal Hase
Hasme Ashutosh	Satish Pagar
Sushil Dwivedi	

5. Analysis

List of Possible causes

Sr. No.	Possible cause
1	Profile Not ok Of Cabin outer panel.
2	Profile not ok of Side Assy at C pillar Area.
3	Wrong Cab Position at C pillar Profile Area of Body shell.
4	Variation in Back Wind Shield Geometry.
5	Wrong Geometry of DC Cab body mtg.
6	Wrong Geometry of Cargo body mtg.
7	Wrong Geometry of Chassis (Cab & Cargo Mounting).
8	Variation in Cab to Cargo Alignment.
9	Poor SOS Adherence for Body Mtg Bolt Fitment at Final line in TCF.
10	Incorrect Profile of Cargo outer panel.
11	SOS Inadequate
12	Untrained Associate on stage.
13	Work content more.
14	Wrong sequence of operation.
15	New Operator Deploy on stage.

Step 5. Analysis

0 - Zero chance
1 - Very low
3- Medium
9 - High

List of Probable causes identification in Brain storming session

Sr no	Possible Causes	Nitin	Prasad	Sudipta	Ashutosh	Patil	Satish	Sushil	Total
1	Profile not ok Of Cabin outer panel.	1	3	1	1	3	1	1	11
2	Wrong Cab Position at C pillar Profile Area of Body shell.	9	9	9	3	3	9	9	51
3	Profile not ok of Side Assy at C pillar Area.	3	3	3	3	3	3	1	19
4	Variation in Back Wind Shield Geometry.	3	1	1	1	3	1	1	11
5	Wrong Geometry of DC Cab body mtg.	3	3	3	3	3	3	3	21
6	Wrong Profile of Cargo outer panel.	3	3	3	3	3	3	3	21
7	Variation in Cab to Cargo Alignment.	1	1	3	3	3	3	3	17
8	Wrong Geometry of Chassis (Cab & Cargo Mounting).	9	3	3	3	3	9	9	39
9	Poor SOS Adherence for Body Mtg Bolt Fitment at Final line in TCF.	0	3	3	1	1	3	3	14
10	Wrong Geometry of Cargo body mtg.	9	3	9	9	9	3	3	45
11	SOS Inadequate	3	3	3	3	3	3	3	21
12	Untrained Associate on stage.	3	3	3	3	3	3	3	21
13	Work content more.	9	9	3	3	9	3	3	39
14	Wrong sequence of operation.	3	3	3	3	3	3	3	21
15	New Operator Deploy on stage.	3	3	3	3	3	3	3	21

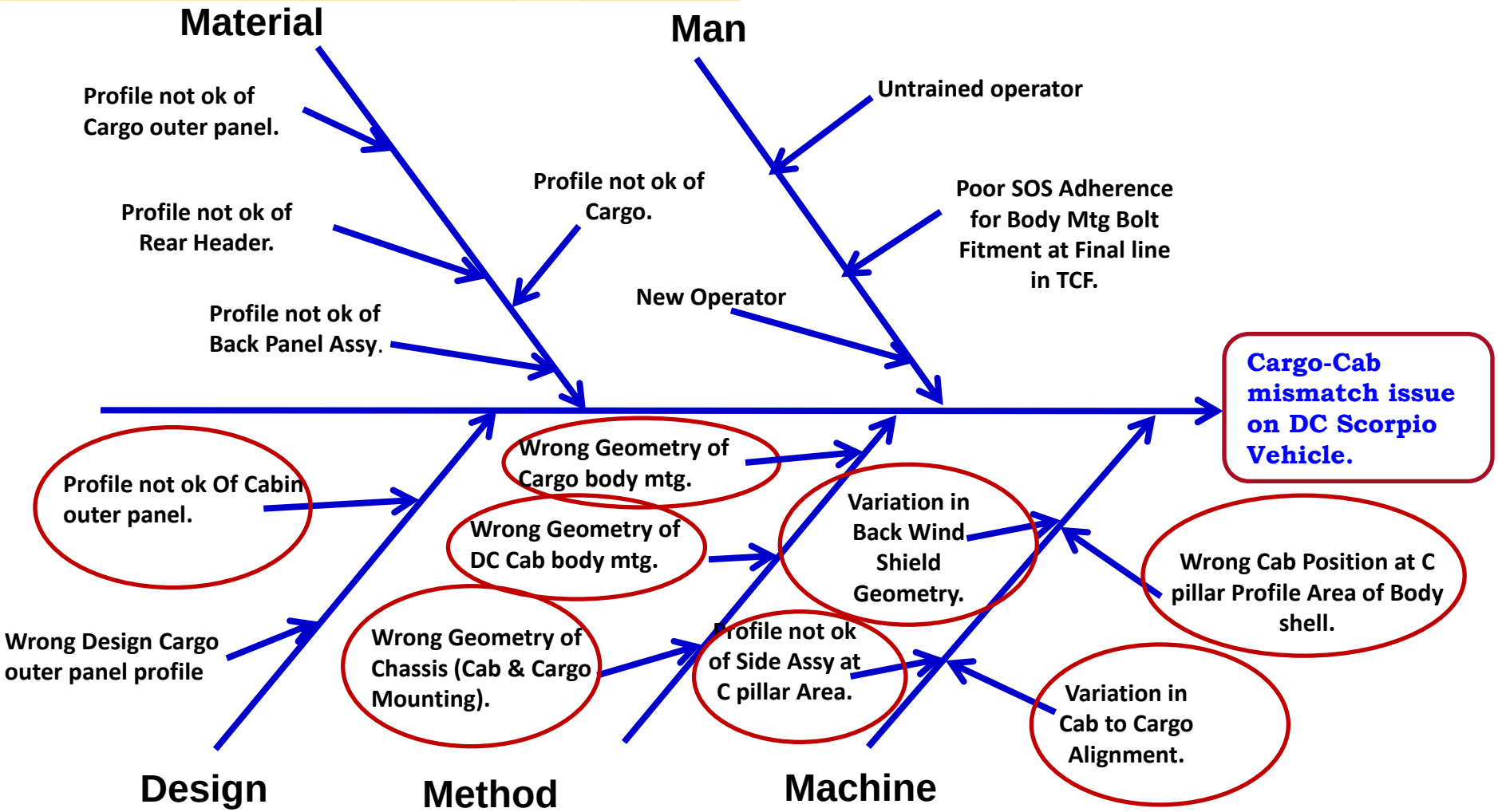
Step 5. Analysis

List of Probable causes identification in Brain storming session

The possible causes who are having the Total score above 30 or more are considered as the probable causes. In Total of 15 Possible causes 3 Probable causes are identified.

5. Analysis

5.1 Identify possible causes:



Note:- All Probable causes are Marked in Fishbone Diagram.

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering

Sr no	Not considered Causes	Reasons
1	Poor SOS Adherence for Body Mtg Bolt Fitment at Final line in TCF.	SOS Followed by associate.
2	Profile not ok Of Cabin outer panel.	Profile Check on Checking Fixture & Found ok.
3	Profile not ok of Side Assy at C pillar Area.	CMM of Side Assy at C pillar Area check & found ok.
4	Variation in Back Wind Shield Geometry.	Back Wind Shield Gauge Reading found ok.
5	Untrained Associate on stage.	All are trained associate.
6	Work content more.	Work content within tact time.
7	SOS Inadequate	SOS is correct.
8	Wrong Geometry of DC Cab body mtg.	CMM report of Body mtg Check & found ok.
9	Profile not ok of Cargo outer panel.	Profile Check on Checking Fixture & Found ok.
10	Variation in Cab to Cargo Alignment.	Total 62 Vehicle Check & found ok.
11	Wrong sequence of operation.	Total 20 Vehicle Check & found ok.
12	New Operator Deploy on stage.	All operators are Skill operators.

5.1 Identify Probable cause

No	List of Probable Causes
1	Profile not ok Of Cabin outer Panel.
2	Profile not ok of Side assy at C pillar Area.
3	Wrong Cab Position at C pillar Profile Area of Body shell.
4	Variation in Back Wind Shield Geometry.
5	Wrong Geometry of DC Cab body mtg.
6	Wrong Geometry of Cargo mtg.
7	Wrong Geometry of Chassis (Cab & Cargo Mounting)
8	Variation in Cab to Cargo Alignment.

5. Analysis

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
1	**Profile not ok Of Cabin outer Panel.**	Profile Inspection done & found ok at Concern area.	**In-valid**

MUSCO STAMPINGS NASHIK

INSPECTION REPORT

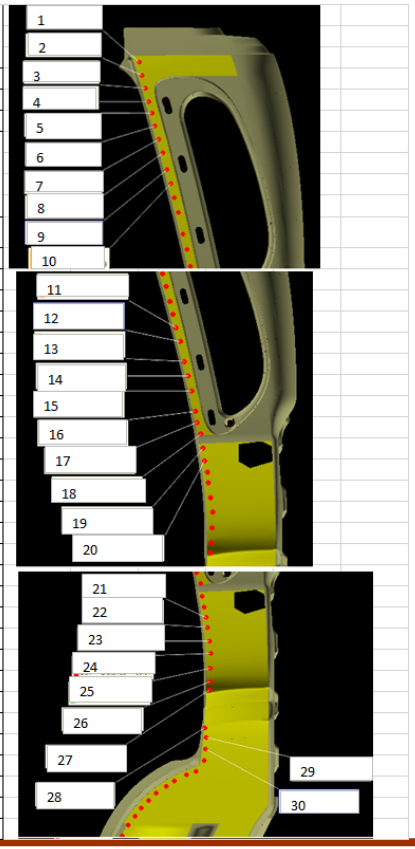
PART NO :- **0101EL0250N** | DATE: **28.06.2017**

PART NAME :- **PNL CABIN OUTER DC LH** | CUSTOMER NAME :- **MAHINDRA AND MAHINDRA**

PREPARED BY :- **CHOUDHARI/KUWAR** | APPROVED BY: NITIN PAWAR

0101EL0250N(LH)

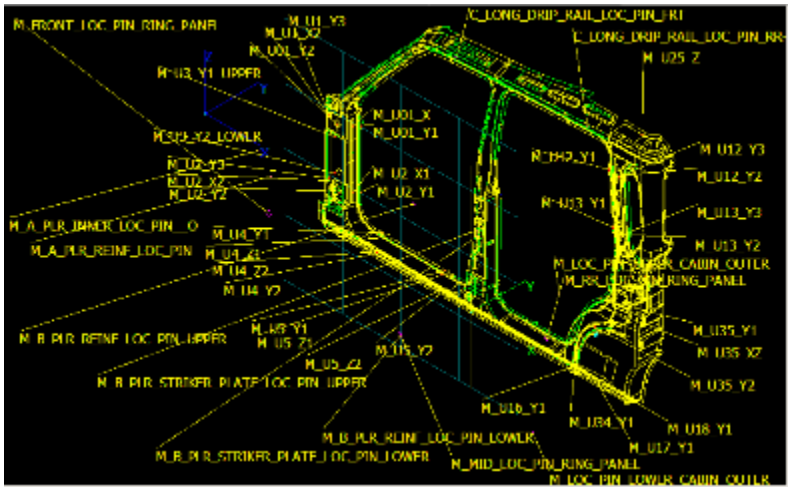
Point	MS	S1	S2	S3	S4	REMARK
1	-0.59	-0.49	-0.51	-0.53	-0.41	OK
2	-0.30	-0.36	-0.30	-0.37	-0.28	OK
3	-0.27	-0.15	-0.10	-0.18	-0.07	OK
4	-0.19	-0.04	-0.05	-0.12	-0.09	OK
5	-0.14	-0.05	-0.18	-0.13	-0.05	OK
6	-0.02	0.03	-0.01	-0.15	-0.11	OK
7	0.08	0.13	0.20	0.11	0.10	OK
8	0.15	0.19	0.18	0.18	0.07	OK
9	0.29	0.28	0.27	0.20	0.23	OK
10	0.39	0.37	0.34	0.29	0.30	OK
11	0.45	0.41	0.39	0.47	0.45	OK
12	0.50	0.41	0.37	0.40	0.52	OK
13	0.48	0.41	0.41	0.39	0.47	OK
14	0.39	0.39	0.35	0.30	0.35	OK
15	0.31	0.36	0.19	0.30	0.27	OK
16	0.32	0.36	0.25	0.25	0.27	OK
17	0.37	0.42	0.34	0.34	0.31	OK
18	0.41	0.51	0.36	0.41	0.47	OK
19	0.62	0.79	0.54	0.63	0.70	OK WRT SAMPLE
20	0.87	0.97	0.67	0.77	0.64	OK WRT SAMPLE
21	0.87	0.97	0.64	0.76	0.79	OK WRT SAMPLE
22	0.78	0.94	0.53	0.68	0.89	OK WRT SAMPLE
23	0.56	0.84	0.50	0.58	0.76	OK WRT SAMPLE
24	0.56	0.75	0.78	0.49	0.60	OK WRT SAMPLE
25	0.51	0.64	0.22	0.36	0.53	OK WRT SAMPLE
26	0.30	0.55	0.59	0.55	0.46	OK
27	0.23	0.47	0.33	0.20	0.34	OK



5. Analysis

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
2	Profile not ok of Side assy at C pillar Area.	Side Assy Profile Check on CMM & found ok.	In-Valid



M_U01_Y1 *(Measured Surface Point , number of points 1)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	1729.372	1729.372	0000.500	-0000.500	0000.000	*******
	Y	-0709.352	-0709.331	0000.500	-0000.500	-0000.020	*******
Center	Z	1146.941	1146.941	0000.500	-0000.500	0000.000	*******

M_U01_Y2 *(Measured Surface Point , number of points 1)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	1705.844	1706.268	0000.500	-0000.500	-0000.424	*******
	Y	-0734.703	-0734.753	0000.500	-0000.500	0000.050	*******
Center	Z	1149.249	1149.249	0000.500	-0000.500	0000.000	*******

M_U03_Y1 *(Measured Surface Point , number of points 1)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	1639.804	1639.812	0000.500	-0000.500	-0000.008	*******
	Y	-0774.226	-0773.857	0000.500	-0000.500	-0000.369	*******
Center	Z	1146.274	1146.274	0000.500	-0000.500	0000.000	*******

M_U2_Y1 *(Measured Surface Point , number of points 1)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	1730.553	1730.545	0000.500	-0000.500	0000.008	*******
	Y	-0708.997	-0709.357	0000.500	-0000.500	0000.360	*******
Center	Z	0774.808	0774.808	0000.500	-0000.500	0000.000	*******

M_U2_Y2 *(Measured Surface Point , number of points 1)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	1646.914	1646.912	0000.500	-0000.500	0000.002	*******
	Y	-0773.916	-0774.014	0000.500	-0000.500	0000.098	*******
Center	Z	0768.975	0768.975	0000.500	-0000.500	0000.000	*******

M_U2_Y3 *(Measured Surface Point , number of points 1)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	1550.414	1550.414	0000.500	-0000.500	0000.000	*******
	Y	-0738.921	-0739.000	0000.500	-0000.500	0000.079	*******
Center	Z	0775.769	0775.769	0000.500	-0000.500	0000.000	*******

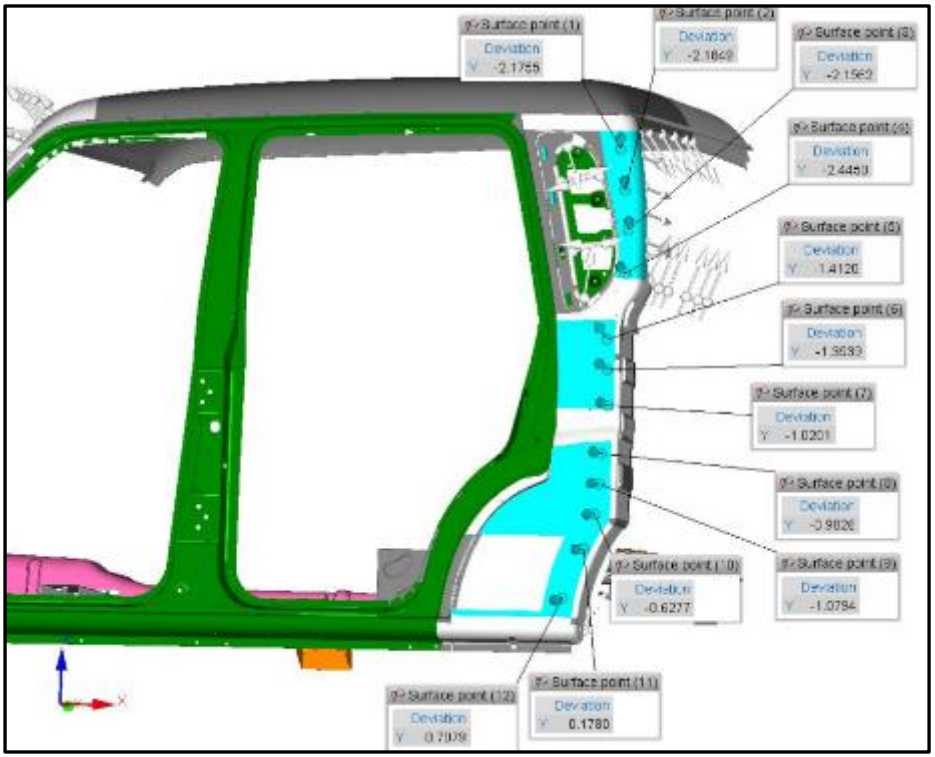
M_U3_Y1_UPPER *(Measured Surface Point , number of points 1)*

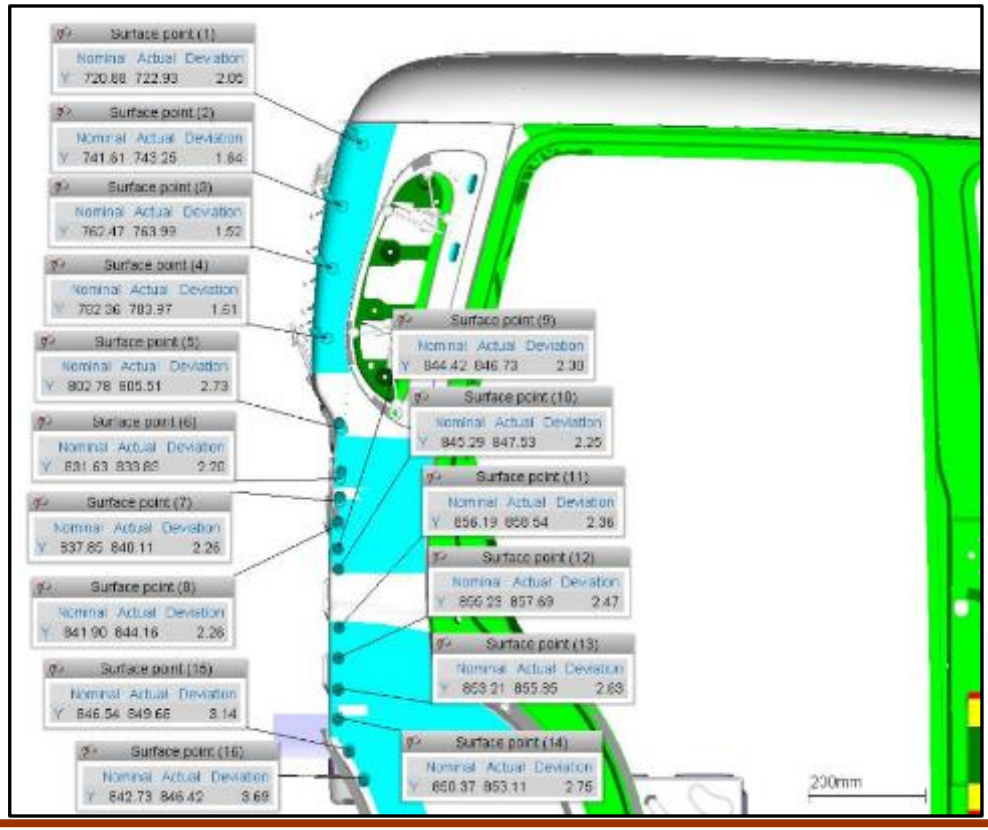
		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	1628.176	1628.176	0000.500	-0000.500	0000.000	*******
	Y	-0780.535	-0781.000	0000.500	-0000.500	0000.465	*******
Center	Z	1073.018	1073.018	0000.500	-0000.500	0000.000	*******

5. Analysis

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
3	**Wrong Cab Position at C pillar Profile Area of Body shell.**	05 BIW Check on CMM at C pillar Profile Area & Found deviated upto3.5mm.	**Valid**

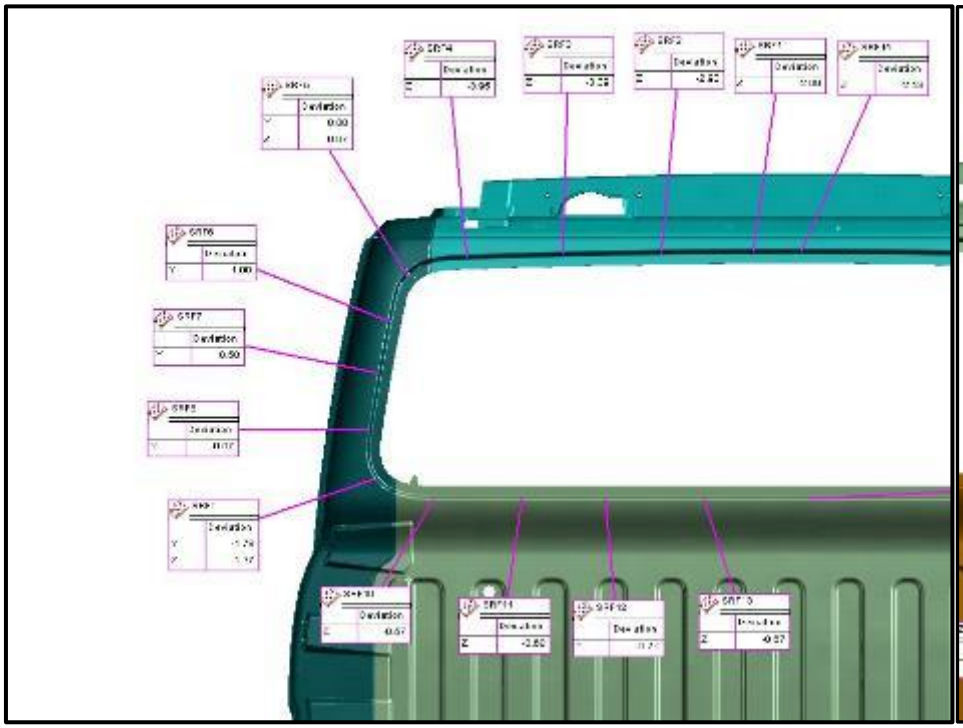


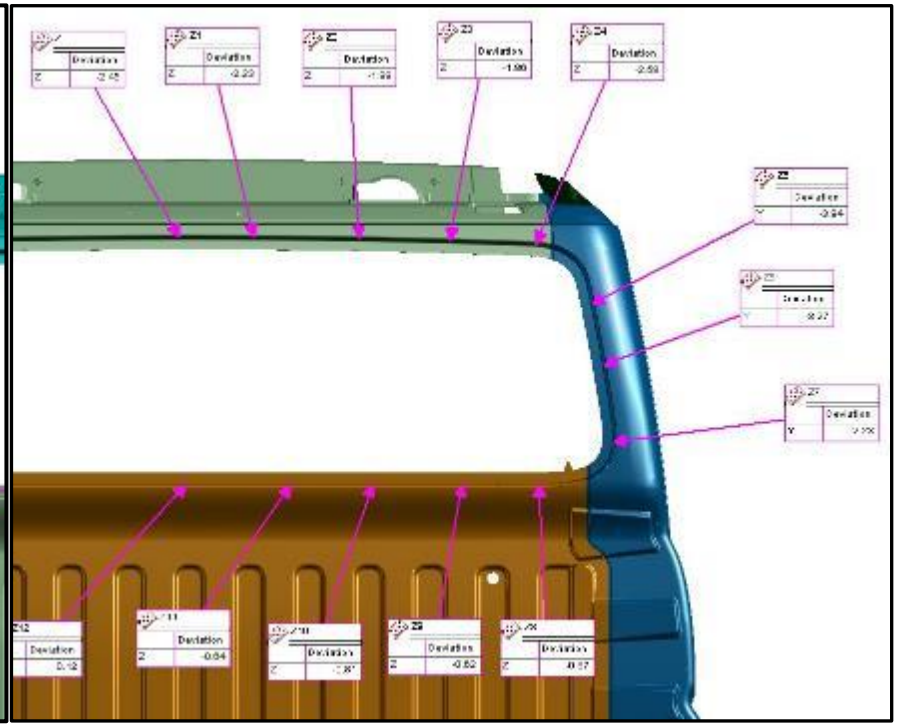


5. Analysis

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
4.	**Variation in Back Wind Shield Geometry.**	1) CMM inspection was done for Back Wind Shield Profile{Qty-05}. 2) Profile co ordinates Found Ok.	**In-Valid**





5. Analysis

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
5.	Wrong Geometry of DC Cab body mtg.	Body mtg Check on CMM & found ok.	In-Valid

SR.	DESCRIPTION OF PART	AXIS	Nominal w109 only	UPR TOL	LWR TOL	DC 15.05.2017
120	BODY MTG HOLE 1 - LH	X	518	1.5	-1.5	-1.41
		Y	-427.5	1.5	-1.5	-1.20
		Z	709.4	1.5	-1.5	-0.85
121	BODY MTG HOLE 1 - RH	X	518	1.5	-1.5	0.11
		Y	-427.5	1.5	-1.5	-1.00
		Z	709.4	1.5	-1.5	-1.52
122	BODY MTG HOLE 2 (LOCATOR PIN ON SILL FRT TRANS) - LH	X	1640	1.5	-1.5	0.52
		Y	-330	1.5	-1.5	0.10
		Z	583	1.5	-1.5	0.62
123	BODY MTG HOLE 2 (LOCATOR PIN ON SILL FRT TRANS) - RH	X	1640	1.5	-1.5	0.36
		Y	-330	1.5	-1.5	-1.31
		Z	583	1.5	-1.5	1.00
124	BODY MTG HOLE 3 - LH	X	2575	1.5	-1.5	-0.08
		Y	-625	1.5	-1.5	-1.40
		Z	583.8	1.5	-1.5	-1.20
125	BODY MTG HOLE 3 - RH	X	2575	1.5	-1.5	0.23
		Y	-625	1.5	-1.5	0.97
		Z	583.8	1.5	-1.5	1.02
126	BODY MTG HOLE 4 - LH	X	3555	1.5	-1.5	0.55
		Y	600	1.5	-1.5	1.20
		Z	722.55	1.5	-1.5	1.50
127	BODY MTG HOLE 4 - RH	X	3555	1.5	-1.5	-0.53
		Y	-600	1.5	-1.5	1.20
		Z	722.55	1.5	-1.5	-1.34

Page 1

5. Analysis

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
6.	**Wrong Geometry of Cargo mtg.**	1. Total 03 Cargo Body mtg Check on CMM at Supplier End. 2. Dimensions are observed out of specification found Not ok.	**Valid**

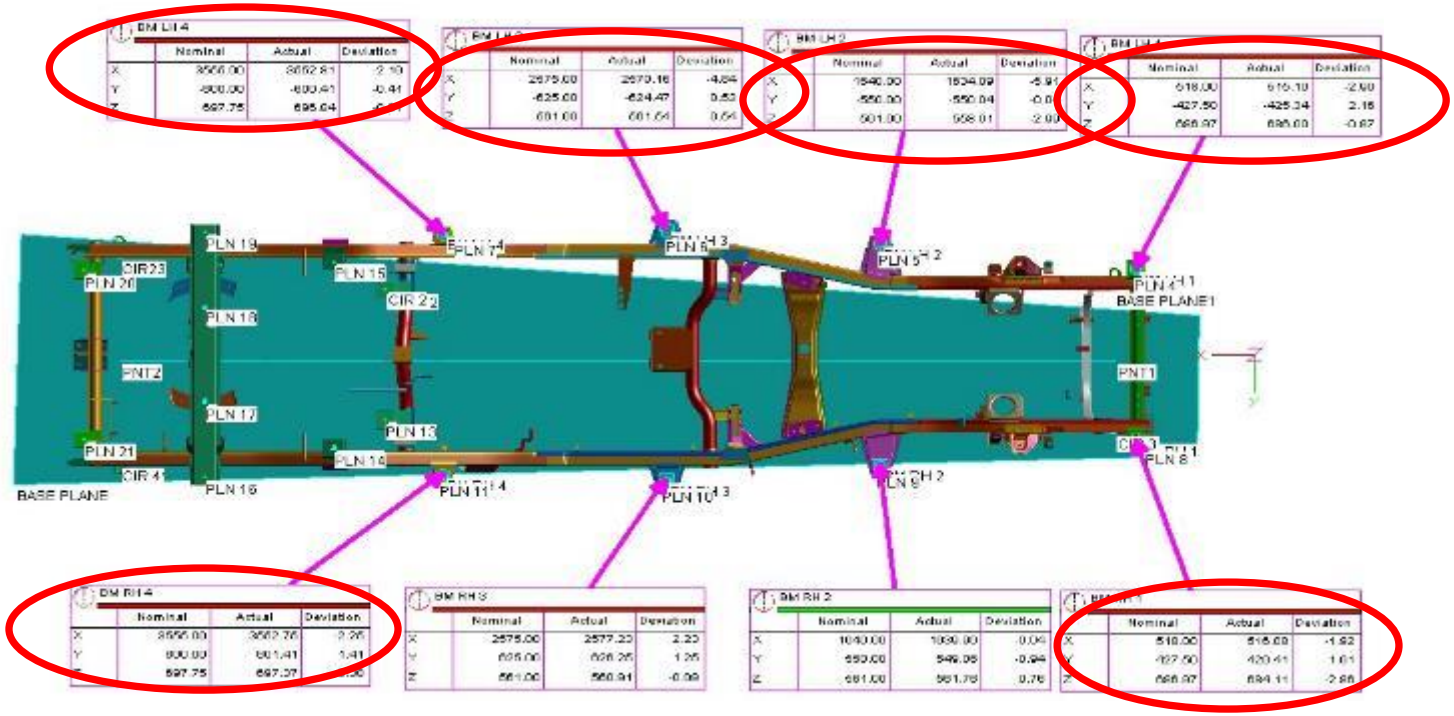
KLT AUTOMOTIVE & TUBULAR PRODUCTS LTD
DIMENSIONAL ANALYSIS REPORT OF SCORPIO-DC
GENERAL TOL IS +/-2.00mm

				FRAME NO.		1789				1803				1810	
SR NO.	DISCREPTION	Nominal	ZDIM NOM	Tol		Actual	Dev	Z ACTUAL DIM	DEV	Actual	Dev	Z ACTUAL DIM	DEV	Actual	Dev
1	OUTRIGGER NO.5A LH														
	X	3422		-1	1	3421.304	-0.696			3420.157	-1.843			3422.299	0.299
	Y	-465		-1	1	-465.679	-0.679			-462.921	2.079			-462.94	2.06
	Z	149.6	8	-2	2	158.142	8.542	6.729	-1.271	155.367	5.767	8.784	0.784	154.729	5.129
2	OUTRIGGER NO.7 CARGO LH														
	X	4519		-1	1	4520.879	1.879			4516.261	-2.739			4520.189	1.189
	Y	-420		-1	1	-416.368	3.632			-416.352	3.648			-415.758	4.242
	Z	157.6	0	-2	2	164.871	7.271	0	0	164.151	6.551	0	0	160.609	3.009
3	BODY OUTRIGGER NO.1 LH														
	X	-95		-1	1	-94.993	0.007			-93.803	1.197			-94.464	0.536
	Y	-427.5		-1	1	-424.595	2.905			-425.958	1.542			-427.772	-0.272
	Z	79	78.6	-2	2	77.877	-1.123	86.994	8.394	78.644	-0.356	85.507	6.907	76.992	-2.008
4	ASSY BODY OUTRIGGER NO.2 LH														
	X	1027		-1	1	1024.074	-2.926			1027.928	0.928			1028.615	1.615
	Y	-550		-1	1	-555.74	-5.74			-550.98	-0.98			-553.402	-3.402
	Z	-47	204.6	-2	2	-39.925	7.075	204.796	0.196	-44.087	2.913	208.238	3.638	-43.036	3.964
5	BODY OUTRIGGER NO.3 LH														
	X	1962		-1	1	1961.567	-0.433			1959.458	-2.542			1960.487	-1.513
	Y	-625		-1	1	-629.377	-4.377			-627.211	-2.211			-627.823	-2.823
	Z	-47	204.6	-2	2	-35.356	11.644	200.227	-4.373	-41.356	5.644	205.507	0.907	-43.886	3.114
6	OUTRIGGER NO.4 LH														
	X	2942		-1	1	2940.632	-1.368			2942.736	0.736			2942.319	0.319
	Y	-600		-1	1	-601.854	-1.854			-600.368	-0.368			-602.008	-2.008
	Z	89	68.6	-2	2	100.352	11.352	64.519	-4.081	99.43	10.43	64.721	-3.879	97.711	8.711
7	OUTRIGGER NO.5 CARGO LH														
	X	3197		-1	1	3199.483	2.483			3197.901	0.901			3198.82	1.82
	Y	-325		-1	1	-325.719	-0.719			-324.999	0.001			-327.08	-2.08
	Z	157.6	0	-2	2	167.216	9.616	-2.345	-2.345	162.703	5.103	1.448	1.448	161.259	3.659
8	OUTRIGGER NO.5 CARGO RH														
	X	3197		-1	1	3197.459	0.459			3195.547	-1.453			3198.7	1.7
	Y	325		-1	1	323.59	-1.41			324.233	-0.767			324.867	-0.133
	Z	157.6	0	-2	2	160.598	2.998	-2.829	-2.829	160.211	2.611	-3.361	-3.361	159.439	1.839
9	OUTRIGGER NO.5A RH														
	X	3422		-1	1	3419.83	-2.17			3421.911	-0.089			3420.184	-1.816
	Y	465		-1	1	464.998	-0.002			467.582	2.582			465.751	0.751
	Z	149.6	8	-2	2	152.692	3.092	5.077	-2.923	152.178	2.578	4.672	-3.328	151.631	2.031
10	OUTRIGGER NO.7 CARGO RH														
	X	4519		-1	1	4516.208	-2.792			4521.863	2.863			4517.965	-1.035
	Y	420		-1	1	421.127	1.127			418.735	-1.265			422.64	2.64
	Z	157.6	0	-2	2	157.769	0.169	0	0	156.85	-0.75	0	0	157.2	-0.4
11	BODY OUTRIGGER NO.1 RH														
	X	-95		-1	1	-96.764	-1.764			-95.613	-0.613			-95.359	-0.359
	Y	427.5		-1	1	428.93	1.43			427.549	0.049			425.73	-1.77
	Z	79	78.6	-2	2	76.046	-2.954	81.723	3.123	75.864	-3.136	80.986	2.386	78.469	-0.531
12	ASSY BODY OUTRIGGER NO.2 RH														
	X	1027		-1	1	1026.988	-0.012			1026.224	-0.776			1023.116	-3.884
	Y	550		-1	1	548.877	-1.123			549.907	-0.093			546.363	-3.637
	Z	-47	204.6	-2	2	-45.877	1.123	203.646	-0.954	-47.51	-0.51	204.36	-0.24	-44.63	2.37
13	BODY OUTRIGGER NO.3 RH														
	X	1962		-1	1	1961.671	-0.329			1961.176	-0.824			1959.204	-2.796
	Y	625		-1	1	623.171	-1.829			625.383	0.383			622.81	-2.19
	Z	-47	204.6	-2	2	-45.685	1.315	203.454	-1.146	-44.672	2.328	201.522	-3.078	-44.672	2.328
14	OUTRIGGER NO.4 RH														
	X	2942		-1	1	2943.94	1.94			2941.158	-0.842			2941.422	-0.578
	Y	600		-1	1	598.392	-1.608			603.202	3.202			599.581	-0.419
	Z	89	68.6	-2	2	95.031	6.031	62.738	-5.862	90.77	1.77	66.08	-2.52	91.976	2.976

5. Analysis

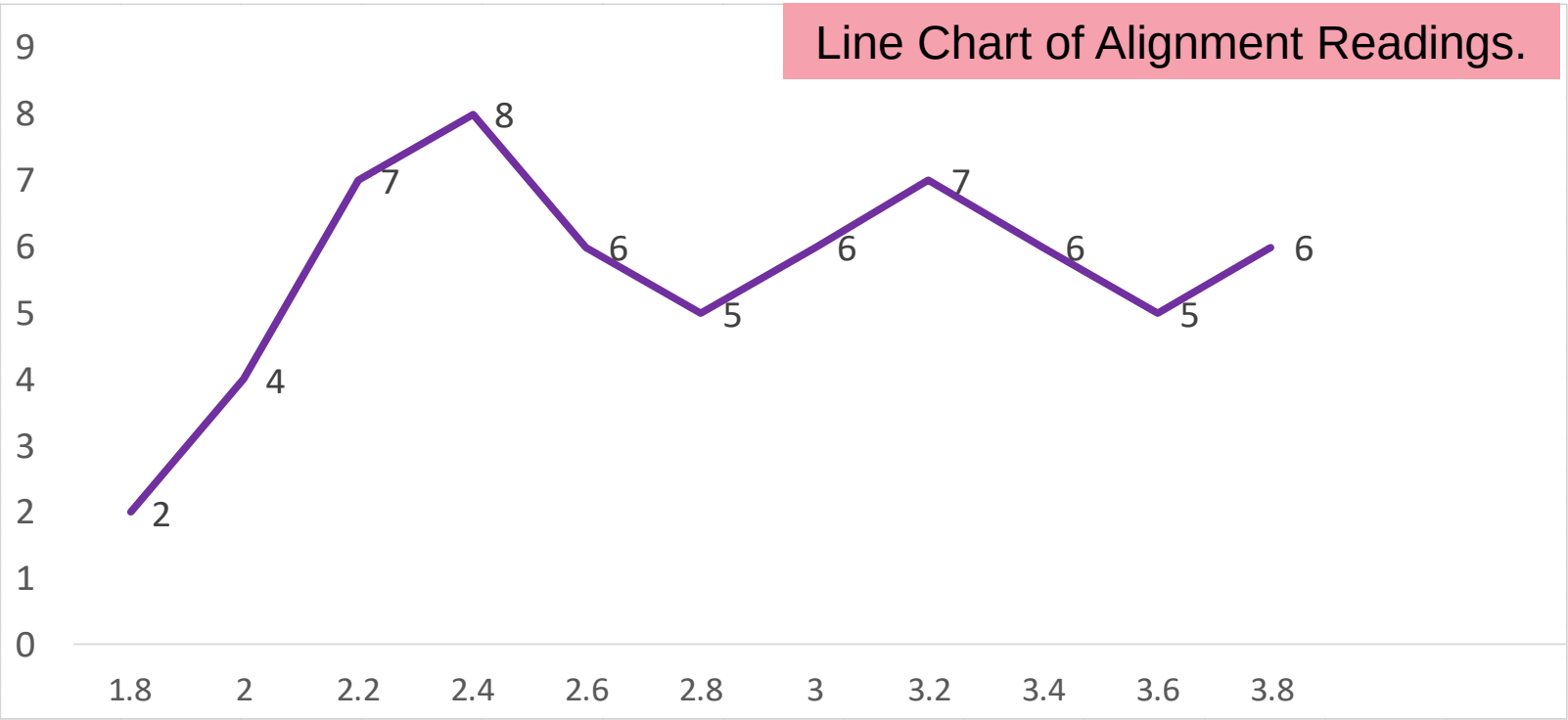
5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
7.	**Wrong Geometry of Chassis (Cab & Cargo Mounting)**	1. DC Chassis Body mtg points Check on CMM at Supplier. 2. Dimensions are observed out of specification found Not ok.	**Valid**



5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
8.	**Variation in Cab to Cargo Alignment.**	1. Total 62 Vehicle Check in TCF. 2. Found Ok. Variation observed from 1.5. to 3.8mm**(Specs. 3.0 ± 1.0mm)**	**In-Valid**



WHY - WHY Analysis

5.3 Identify the Root Causes:

Sr No	Probable cause	why	why	why	why	ACTION / COUNTERMEASURE
1	**Framing Fixture Stn#20,,**Wrong Cab Position at C pillar Profile Area of Body shell.	Side Assy Position is deviated in Y-direction.	Framing Welding Fixture not ok.	Improper locating & clamping of Side assy at C pillar Area.	Locating pins & resting mylers position not ok.	**Framing Fixture Stn#20,Locating pins & resting mylers position Rectified.**
2	Wrong Geometry of Cargo mtg.	Cargo Body mtg Trigger are out of specification.	Cargo Welding Fixture not ok.	Improper Resting in Y-direction & Clamp Gap for Body mtg Trigger.	Resting mylers & Clamp position not ok.	**Cargo Assy Stn#10, Resting mylers & Clamp position Adjusted.**
3	Wrong Geometry of Chassis (Cab & Cargo Mounting)	Chassis Body mtg bkts are out of specification.	Chassis Welding Fixture not ok.	Improper Resting in Y-direction & Holding for Body mtg bkt.	Resting mylers & Clamp holding position not ok.	**Chassis Assy Stn #10 & #30 ,Resting mylers & Clamp holding position fixed.**

6. Implement the corrective measures

Sr No	Root Causes	Action plan	Target Date	Status
1	Incorrect Cab Position at C pillar Profile Area.	1. FARO-ARM CMM Inspection done for Framing Welding Fixture. 2. Framing Welding Fixture Stn#20, Locating pins & resting mylers position are rectified.	20.06.17	Completed

Framing Fixture Report

M_Y1_UNIT-05 *(Measured Surface Point , number of points 1)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	2300.96	2300.96	0000.50	-0000.50	0000.00	*******
	Y	-0819.81	-0819.78	0000.50	-0000.50	-0000.03	*******
Center	Z	0585.46	0585.47	0000.50	-0000.50	-0000.01	*******

M_Y2_UNIT-05 *(Measured Surface Point , number of points 1)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	2298.87	2298.86	0000.50	-0000.50	0000.00	*******
	Y	-0732.10	-0732.24	0000.50	-0000.50	0000.15	*******
Center	Z	0635.45	0635.50	0000.50	-0000.50	-0000.05	*******

M_Y1_UNIT-04 *(Measured Surface Point , number of points 1)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	1889.07	1889.07	0000.50	-0000.50	0000.00	*******
	Y	-0818.29	-0818.19	0000.50	-0000.50	-0000.10	*******
Center	Z	0585.97	0585.99	0000.50	-0000.50	-0000.02	*******

M_Y2_UNIT-04 *(Measured Surface Point , number of points 1)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	1889.22	1889.22	0000.50	-0000.50	0000.00	*******
	Y	-0725.35	-0725.41	0000.50	-0000.50	0000.06	*******
Center	Z	0647.37	0647.05	0000.50	-0000.50	0000.33	*******

M_Y1_UNIT-02 *(Measured Surface Point , number of points 1)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	1673.75	1673.75	0000.50	-0000.50	0000.00	*******
	Y	-0841.18	-0840.79	0000.50	-0000.50	-0000.39	*******
Center	Z	0773.31	0773.35	0000.50	-0000.50	-0000.04	*******

6. Implement the corrective measures

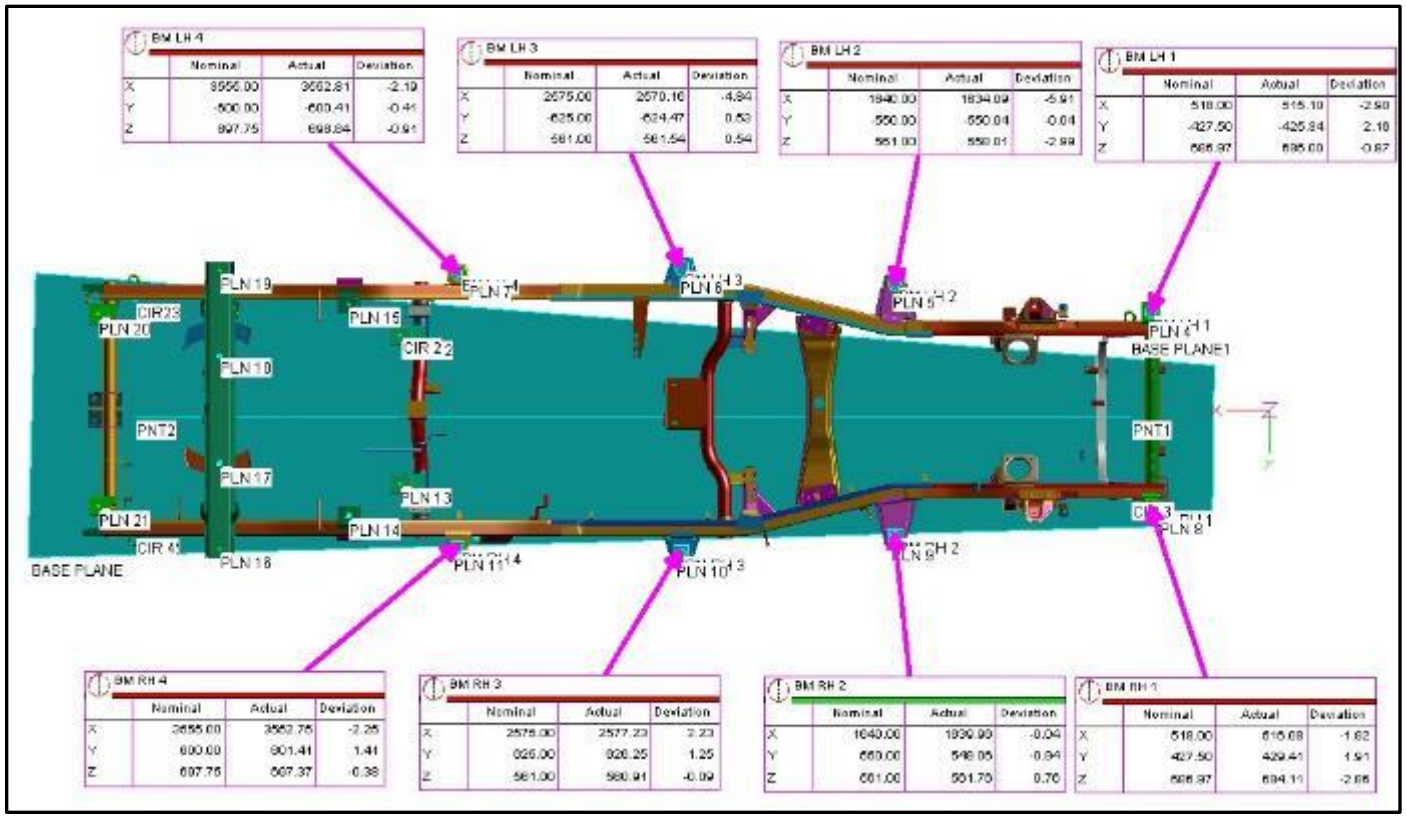
Sr No	Root Causes	Action plan	Target Date	Status
2	Incorrect Geometry of Cargo mtg.	Cargo Welding Fixture#10 Resting mylers & Clamp position adjusted.	05.07.17	Completed

Cargo Body Mtg bracket position shift by adding 1 mm shim at Y-direction.

Cargo Body mounting bracket position shift by adding 1 mm shim at Y-direction.

6. Implement the corrective measures

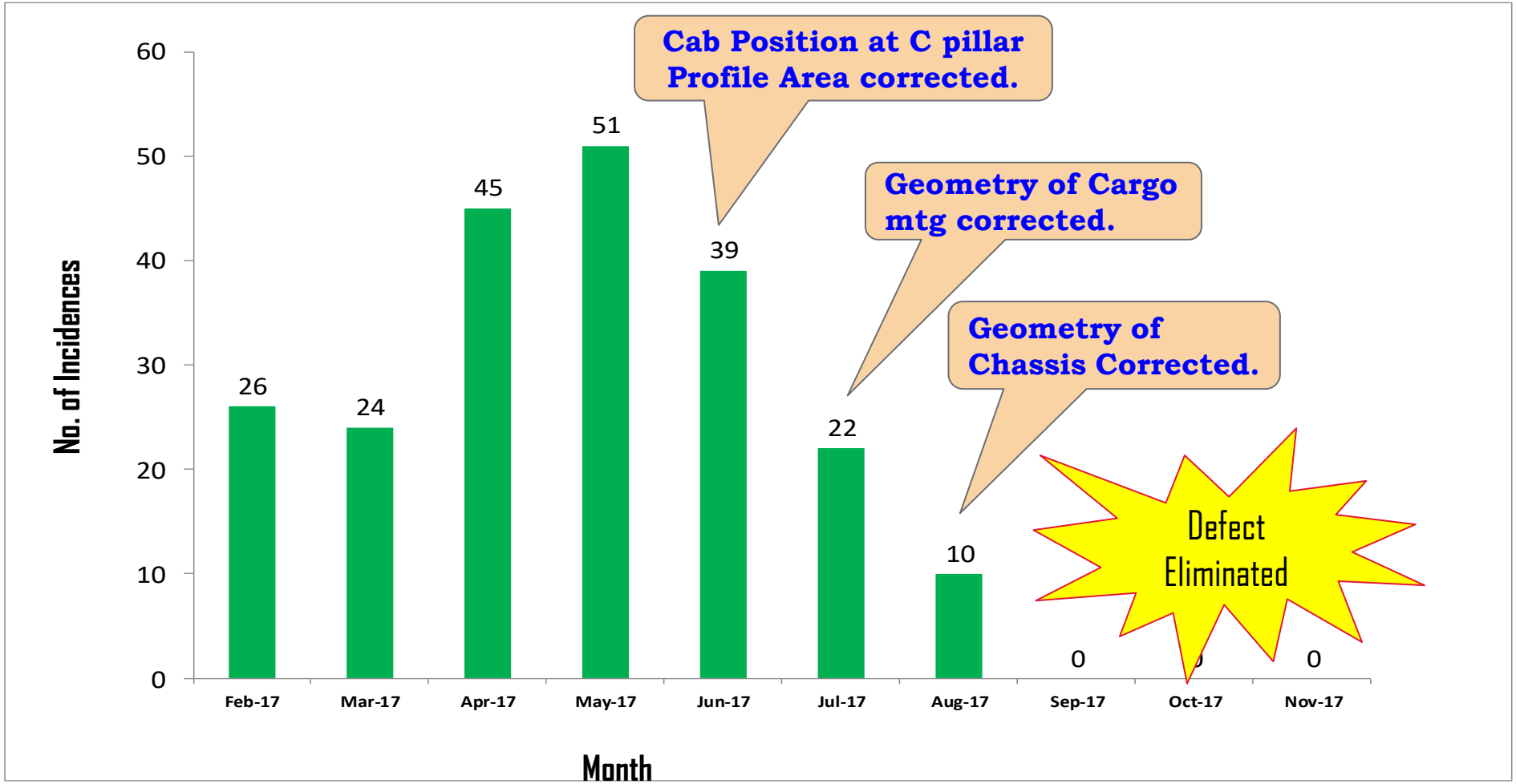
Sr No	Root Causes	Action plan	Target Date	Status
3	Geometry of Chassis for Cab & Cargo Mounting.	Chassis Welding Fixture Stn#10 & 30, Resting mylers & Clamp holding position fixed as per CMM report.	11.08.17	Completed



CMM Report of Chassis Fixture.

7. Check & confirm the results & effects

7.1 Verify Results:



7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

	Activity	Target Dt	Resp.	Status
1	Framing Welding Fixture Locating pins & Mylers Correction done as per CMM report.	20.06.17	Sushil/Nitin	Closed
2	Cargo Welding Fixture#10 corrected as per CMM report in Resting mylers to improve repeatability.	05.07.17	Bagade U.	Closed
3	Chassis Welding Fixture Check against Cad Data & corrections as per CMM Report.	11.09.17	Patil	Closed
4	Fixture History card of Framing Fixture has updated.	20.06.17	Bagade U.	Closed
7	Fixture History card of Cargo Welding Fixture has updated.	05.07.17	Sushil	Closed
5	OPL Given to all the stage operators & On Job Training given to all the stage operators with the I Do, We Do and You Do.	15.10.17	Ganesh	Closed
6	PFMEA Updation of Framing Fixture.	20.06.17	Sushil/Nitin	Closed

7.3 Tangible benefits:

- **ICC PPM improved by 3.85% i.e. from 45042 to 43265 per month.**
- **Improvement in STR by 0.9% (Straight Through ratio)**
- **01 Field concern (Chile) & 95 incidences CAI Concern of Cab & Cargo Mismatch reduced to Zero from the month of April-17 to 20th Nov-17.**
- **s 72008 Rupees/year.**

7.3 Intangible benefits:

- **Customer Satisfaction Increased Due to Zero concern.**
- **QC Story approach studied and used for concern resolution.**
- **Stage Operators Moral Improved.**
- **Team Work Built up.**

7.3 Tangible benefits:

Cost Saving by Eliminate Quality Concern of Cab & Cargo Mismatch Issue on DC Scorpio Vehicle. - Message (HTML)

FILE MESSAGE

Ignore Junk Delete | Reply Reply All Forward Meeting IM More | Move to: ? Team Email Reply & Delete To Manager Done Create New | Move Rules OneNote Actions | Assign Policy Mark Unread Categorize Follow Up | Translate Find Related Select | Zoom

Delete Respond Quick Steps Move Tags Editing Zoom

Wed 06/12/17 3:51 PM

AMRUTKAR GOKUL

Cost Saving by Eliminate Quality Concern of Cab & Cargo Mismatch Issue on DC Scorpio Vehicle.

To HIRAY NITIN

Dear Nitin ,

Please find below details related to cost of your MYB project .

Description	Feb-17 to April-17
Number of pieces rejected last month (for the part number identified for study)	95
Number of pieces(Vehicle) reworked last month	95
Rework man hours Cost Including (Labor Cost+ FOS+ Electricity) in SAP.	758 ` / hr.
Time required to rework the defect.	15 min Avg.
Total time required to rework the defects (in mins.)	1425 Mins.
Total time required to rework the defects (in Hrs.)	23.75 Hrs.
Total rework cost(Rs.) for Feb-17 to April-17 month	18002.50 `

Regards,

Gokul

8.1 Updation of Document

S N	Document	Actual date	Responsibility	Status
1	**Fixture History card of Framing Fixture has updated**	**20.06.17**	**Sushil**	**Closed**
2.	**Fixture History card of Cargo Assy Stn#10 Fixture has updated.**	**05.07.17**	**Uday**	**Closed**
3.	**PFMEA Updation for Framing Fixture & Cargo Fitment Stage.**	**20.06.17**	**Nitin**	**Closed**

8.2 Training.

Not Applicable.

8.3 Additional Audit or Check

1	**Checkpoint is added in IQ check sheet**	**01.10.17**	**Satish**	**Closed**

9. Summarize & Plan Future actions

9.1) Conclusion :

Cargo-Cab mismatch issue on DC Scorpio Vehicle has eliminated by taking action on all the Probable causes of identification at In-house & Supplier End.

9.2) Horizontal Deployment :

- **Check the Possibility of Extending the actions to similar for BMT model in Nashik & Chakan Plant.(02nd Dec-17).**

9.3) Lessons Learnt :

Things Gone Wrong : Inappropriate Welding Assy Fixtures Layout & Lack of proper Interface between BIW Geometry, Cargo Geometry & TCF inter Face.

Things Gone Right : We studied & Understand the TCF interface with Welding & Fitment Fixture & implemented the in-house actions.

Thank you

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